

Nuclei Extraction Protocol

Keep everything cold and move as quickly as is safely possible.

Required Buffers:

Extraction Buffer*: Dissociation Buffer + 1% Kollidon VA64 + 1% Triton X100+ 1:40
RNase-inhibitor

Wash Buffer: Dissociation Buffer (DB)

Prior to extraction:

Set up all needed buffers and equipment so you can move as smoothly and quickly as possible through the extraction. Below is an example of what is needed for the extraction of 3 samples (prior to chilling):

For each sample you wish to extract you will need the following:

- Clean glass slide and new razor blades, at 4C
- Dissecting tray, kept at -20C
- Ruler
- 3 mL syringe with 26 ½ gauge needle, at 4C (have an extra chilled one on hand in case the first one clogs)
- 2 mL chilled extraction buffer, placed in the first well of 12-well plate
- 6-well tissue-culture plate, well-bottoms colored with marker, on ice
- ~50 mL chilled DB
- DAPI (1:1000) (<https://www.thermofisher.com/order/catalog/product/D1306>)
- 100um filter
(<https://us.vwr.com/store/product/14796270/vwr-cell-strainers-dnase-rnase-free-n-on-pyrogenic-sterile>)
- 50 µm eppendorf tube filter:
(<https://us.sysmex-flowcytometry.com/consumables/celltrics-filters/sterile-single-pack-celltrics-filters/1445/sterile-single-pack-celltrics-filters-50/box?number=04-004-2327>)
- 4 50 mL tubes, on ice
- c-Chip FR hemacytometer

For FACS:

- 96 well cold block:
https://www.daigger.com/eppendorf-pcr-coolers-14616-group?gclid=EAlaIQobChMI2rKM7aDN4AIVDRgMCh2jkwRMEAQYBSABEgKeA_D_BwE
- 200uL 5% BSA-DB
- 20uL collection buffer: DB with 1:40 RNase-inhibitor

Extraction:

1) Retrieve dissection tray from -20C and the glass slide and razor blade from 4C.

2) Place the glass slide, razor blades, and ruler on the dissection tray and perform all cutting on slide to minimize damage to, and contamination of, the tray.

3) Place the frozen sample on the glass slide and shave the frosty/freezer-burned tissue face you want to sample from off the sample and discard. This is damaged and not worth extracting. Then, moving rapidly but carefully, shave off “enough” sample to work with. The photo below illustrates approximately how much to remove. Exact measurements are not necessary. If blood or obvious white matter is present in the shaved sample, try to remove it and discard.



4) Return the tissue block to -80C (or put on dry ice as a temporary hold).

5) Rapidly mince the shaved sample with the 2 chilled razor blades that was used to shave the tissue block and scrape into the first well of a 6-well plate that contains 1.5 mL of cold extraction buffer. Pipette up and down 20 times.

6) Commence 10-min incubation, pipetting up and down 20x every ~2.5 min with a 1000 uL pipette.

7) Pull the sample into the chilled needle/syringe and express into the same well

8) Repeat this, pulling the sample into the same syringe and expressing it into the same well

9) (This step may not be necessary if you do not see a lot of white matter in your samples) Wet the 100- μ filter with 1000 uL DB then run sample through the filter into a 50 mL falcon tube

11) Put 1mL chilled DB through the filter to help any nuclei trapped in the filter go through.

12) Add chilled DB to the extracted nuclei to a total volume of just under 30 mL.

13) Divide these diluted nuclei among 2 50 mL conical tubes (~14 mL per tube).

14) Spin at 500 rcf for nuclei for 10 minutes at 4C.

15) Carefully remove as much of the supernatant as possible without disturbing the pellet (which is typically impossible to see – the cleaner the prep, the less there is to see).

16) Pipette up and down to resuspend nuclei in the remaining ~200-250ul in each of the 2 tubes. Combine all of the nuclei suspensions (~0.5ml) into 1 tube (Eppendorf)

17) Wet the **50um eppendorf tube filter with DB**, pass the 1mL of washed sample through the filter.

17) Add **1ul of DAPI** to 1ml of nuclei. Remove 20ul and place into FR hemocytometer.

18) Visualize with the fluorescent scope, getting images of brightfield and DAPI-excited nuclei. It is normal to see some debris.

19) If sample shows good nuclei, then go to FACS and gate based on DAPI/Hoechst fluorescence.

20) Preparing for FACS:

We collect samples in PCR tubes held in a chilled (-20) 96-well cold block to keep the collection volume as minimal as possible:

https://www.daigger.com/eppendorf-pcr-coolers-14616-group?gclid=EAlaIQobChMI2rKM7aDN4AIVDRgMCh2jkwRMEAQYBSABEgKeA_D_BwE

- Prepare the PCR tubes:
 - Fill the PCR tube with 5% BSA-DB, vortex, and remove BSA. This coats the sides of the tube so that if nuclei hit the side of the tube they will fall to the bottom and not dry out.
 - Wash tube with 200 uL DB
 - Make collection buffer, add 20uL collection buffer to the PCR tube.

FACS into the tube using a 70um chip with 1%FSC threshold on the DAPI peak

20) After FACS, use a pipette to determine the volume in each tube

23) Take the concentration: combine 18ul chilled DB with 2ul of nuclei into Eppendorf to make a 1:10 dilution. Mix and put into FR hemocytometer.

24) Visualize with the fluorescent scope, getting images of brightfield and DAPI-excited nuclei. Notice that debris is gone.

25) Count 3 larger squares and calculate the average of the 3 squares. Multiply that number by 10 (accounts for 1:10 dilution) then multiply by 5 (factor to account for hemocytometer). This number is your final concentration!

26) Look at 10x chart to determine what concentration is needed for the amount of nuclei that you want to load into the chip. Based on that, you can always dilute your nuclei more to load a lower concentration.